

ORIGINAL ARTICLE

Application of Convolutional Neural Network Image Analysis and Machine Learning to Basic Blood Tests for Intelligent Diagnostic Assistance

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ABSTRACT

Background and Objectives: We developed an automated morphological image recognition deep learning system (image recognition DLS) of peripheral blood cells, then constructed the diagnostic assist DLS combining image recognition DLS data with complete blood count (CBC) data. This study aimed to evaluate the clinical performance of the image recognition DLS and the diagnostic assist DLS in routine examinations.

Methods: The image recognition DLS was trained using datasets containing 1476727 images of white blood cells (WBCs), nucleated red blood cells (NRBCs), and large platelets to differentiate 14 blood cell types and to recognize 24 morphological characteristics. CBC data were obtained through the automated hematology analyzer (Sysmex XN-9000) and combined with the image recognition DLS data to construct the diagnostic assist DLS. The clinical performance of the image recognition DLS was evaluated using 128716 blood cell images from 589 smears obtained from healthy subjects, ALL, AML, ML, MPN, and MDS cases in routine examinations.

Results: The image recognition DLS classified 14 blood cell types with an accuracy of 97.3%–99.9%. The accuracy of 11 morphological characteristics exceeded 90%. Blast cells were detected accurately on all slides, where they were identified by manual microscopy. Malignant lymphocytes were classified as blasts and/or lymphocytes with the morphological characteristics of each subtype of lymphoma. The diagnostic assist DLS successfully differentiated MDS, achieving an AUC (area under the curve) of 0.99.

Conclusion: This study demonstrated the potential of the diagnostic assist DLS, utilizing morphological image recognition DLS data combined with CBC parameters, as a promising diagnostic tool.

Yuki Horiuchi and Mendamar Ravzanaadii contributed equally to this study.

1 | Introduction

Accurate identification of peripheral blood (PB) cells using microscopic examinations is an essential step in diagnosing hematological disorders. Morphologically identifying leukemic and malignant lymphoma (ML) cells can serve as a powerful tool for diagnosing these diseases [1, 2].

Since microscopic examinations of the PB smears are tedious, and the accuracy depends on the examiners' skills, automated hematology analyzers have been widely used to count cell numbers and differentiate blood cell types. However, due to the detection algorithms, there is a limitation to identifying malignant cells by these analyzers [3].

Therefore, automated cell morphology classification systems using artificial intelligence (AI) have been developed, and they are sufficiently accurate in differentiating normal samples [4, 5]. However, the current systems are not accurate enough to detect abnormal cells and diagnose hematopoietic cancers [5, 6]. In 2019, ICSH published recommendations on the application of digital imaging technology. They stated the pre-classification of blood cells using AI algorithms is useful [7], but the more accurate AI image analysis systems are awaited [8].

In the medical field, image diagnosis supports the utilization of convolutional neural networks (CNNs), which learn from a large number of medical images, to prevent diagnostic errors, delays, and standardize medical care [9]. However, it has been noted that it is challenging to provide a sufficient number of labeled images for training deep learning models, and the time-cost of experts providing basic annotations is a major factor affecting model learning [10].

We previously developed a deep learning system (DLS) for image recognition, specifically a CNN-based DLS to differentiate leukocytes [6]. The DLS consists of interconnected multilayer nodes, with each layer enhancing and optimizing predictions and classifications based on the preceding layer. The input layer integrates data preprocessing for the model, whereas the output layer provides the final prediction or classification. Our DLS was constructed using a dataset containing 695 030 blood cell images, including hematologic malignant cells, and has demonstrated that it can differentiate myelodysplastic syndrome (MDS) from aplastic anemia in cases of pancytopenia [6]. We also demonstrated that combining image recognition DLS data with complete blood count (CBC) data from the automated hematology analyzer enables differentiation of polycythemia vera, essential thrombocythemia, and myelofibrosis in Ph-negative myeloproliferative neoplasm (MPN) cases. This showcases its potential as a diagnostic assist DLS [11].

In this study, we aimed to evaluate the clinical performance of both image recognition DLS and diagnostic assist DLS in routine examinations. To achieve this objective, the image recognition DLS was reconstructed using a training set of 1,476,727 images. Then, we assessed the clinical performance of the DLS in identifying abnormal samples, utilizing 589 PB smears obtained

from routine examinations at a university hospital in Japan. Our study demonstrated: (1) the effectiveness of the image recognition DLS in cell classification and detection of morphological abnormalities, and (2) the performance of the diagnostic assist DLS in discriminating hematopoietic neoplastic diseases such as leukemia, ML, and MDS.

2 | Materials and Methods

2.1 | Dataset Characteristics and Disease Classification

A total of 128 716 images from 589 PB samples of 214 healthy control (HC) subjects and 273 patients at Juntendo University Hospital (Chiyoda-ku, Tokyo) from 2017 to 2021 were used for evaluation of the image recognition DLS. The HC group was defined based on the reference ranges of the Japanese Committee Clinical Laboratory Standards: $3.5 \times 10^9/L \leq$ white blood cell (WBC) $\leq 9.0 \times 10^9/L$, $110 \text{ g/L} \leq$ hemoglobin (Hb) $\leq 170 \text{ g/L}$, $150 \times 10^9/L \leq$ platelets (PLT) $\leq 400 \times 10^9/L$; immature granulocytes (promyelocytes + myelocytes + metamyelocytes) $\leq 2\%$; no blasts, nucleated red blood cells (NRBCs), or megakaryocytes present on PB smear; no apparent abnormal leukocyte fraction; high-sensitivity C-reactive protein (hs-CRP) $\leq 0.3 \text{ mg/L}$; and no obvious underlying diseases such as malignancy, active infection, or collagen disease [12, 13]. The disease classifications included acute lymphoblastic leukemia (ALL, 26 smears from 13 patients), acute myeloid leukemia (AML, 52 smears from 19 patients), ML (84 smears from 59 patients), MDS (with suspected cases, 97 smears from 91 patients), MPN (50 smears from 33 patients), and other hematopoietic neoplasms and related disorders such as aplastic anemia, primary immune thrombocytopenia, and cytopenia without hematopoietic malignancies, non-hematologic malignancies, and collagen diseases (66 smears from 58 patients).

The PB was collected using K2EDTA anticoagulation tubes (Terumo, Tokyo, Japan). The PB smears were prepared and stained with May-Grunwald-Giemsa using fully automated smear makers (Sysmex SP-50 and SP-1000i). Subsequently, digitalized cell images were collected with the FDA-approved DI-60 automated digital cell image analyzer (Sysmex, Kobe, Japan). The digital blood cell images satisfied all applicable ICSH requirements for blood cell morphology assessment before being analyzed by the DLS [7].

In parallel, as a reference method for evaluating the performance of the DLS, scanned images were independently reviewed and annotated by three laboratory technologists board-certified in hematology, along with one senior hematopathologist who is different from the ones labeling the training datasets. They used the morphological criteria of the CLSI guideline, blood cell atlases, and WHO classification of myeloid neoplasms and acute leukemia [14–22]. For smudge cell classification, if the cytoplasm is destroyed but the nucleus remains intact and the cytoplasmic content is still visible, the cell is classified as its original type with a morphological characteristic of a smudge cell. However, if only amorphous nuclear debris is observed, the cell is classified as a smudge cell.

2.2 | Construction of the Image Recognition DLS and Its Validation

The cell image recognition DLS was upgraded from our previously developed CNN-based system using Keras and TensorFlow. The DLS consists of a feature extraction module composed of two submodules. The first submodule comprises three consecutive blocks, each containing two parallel convolutional pathways designed to optimize feature extraction from input cell images. The second submodule includes eight consecutive blocks, also with two parallel paths: one composed of convolutional layers and the other acting as a residual (non-convolutional) bypass to prevent activation saturation and facilitate gradient flow. Each convolutional path utilizes Separable Convolution (a depthwise separable variant of Conv2D), followed by Batch Normalization (BN) and a Rectified Linear Unit (ReLU) activation function [6].

The DLS was designed to prioritize stable morphological features, including nuclear shape, cytoplasmic texture, and cell size, which remain consistent across different staining protocols and imaging conditions, making them suitable for robust model training [23]. To further reduce the risk of over-learning and ensure stable performance during updates, the architecture incorporates residual network modules and convolutional layer bypass pathways to prevent deep layer saturation and reduce sensitivity to imaging noise and artifacts. Additionally, batch normalization layers were implemented to normalize input data distribution, thereby enhancing training stability and generalization. These architectural strategies collectively help maintain consistent performance and minimize overfitting, even when the model is retrained with new image data.

Training datasets comprising 1476727 normal and abnormal cell images were used to enable pattern recognition. The data were categorized into 14 cell types, including WBCs, NRBCs, and platelets (Figure 1A), along with associated abnormal morphological characteristics identified by the DLS (Figure 1B). The number of images per cell type ranged from 2002 to 690181, and per abnormal morphological characteristics from 1065 to 49871. The validation dataset included 12403 images, with 138–2221 images per cell type and 52–218 per morphological characteristic (Table S1).

The image recognition DLS provides two types of output: cell classification and morphological characteristic information. For each smear, cell classification results are reported as the rate of occurrence per cell type, based on the DLS's inference. The system assigns an inferential value (0%–100%) to each of the 14 cell types, indicating the likelihood of that type being present, and outputs the one with the highest score. For abnormal morphological characteristics, the DLS calculates their frequency within each cell category and assigns an inferential value to each morphological characteristic. Only morphological characteristics exceeding a predefined threshold are presented as part of the final output.

We used the validation data to generate ROC (receiver operating characteristic) curves for each morphological characteristic by JMP (John's Macintosh Project) Pro 17 and determined the threshold as the inferred value with the highest sensitivity and specificity. A total of 24 morphological characteristics were selected based on the following criteria: cell lineage dysplasia in MDS, an adequate number of images for validation (> 40), high

AUC (area under the curve) (> 0.8), and high sensitivity and specificity. However, two morphological characteristics (plasma cell-like and flower cell) were included due to their unique appearance in mature lymphoid neoplasms, despite not meeting all the criteria (Figure 1B). The DLS can detect 200 blood cell images on a PB smear within a few seconds.

The validation results of the DLS's performance in cell classification and identification of morphological characteristics are shown in Table S1. The DLS displayed excellent sensitivity and specificity, both exceeding 90% for all blood cell types. Cohen's *h* values were below 0.5 in all cell types, indicating a small effect size in performance variability. The sensitivity and specificity of morphological characteristics varied, ranging from 48.6% to 100% and from 71.2% to 98.9%, respectively. The AUCs for detecting morphological characteristics in band neutrophils, segmented neutrophils, and lymphocytes were all above 0.90, except for the Döhle body detection in segmented neutrophils and the identification of giant platelets.

2.3 | Construction of the Diagnostic Assist DLS

A diagnostic assist DLS, based on a random forest (RF) classifier, was designed to differentiate between HC, MDS, and non-MDS conditions (ALL, AML, ML, MPN, and other diseases). Although various machine learning algorithms are available for classification tasks, including XGBoost, which was used in our 2019 study, RF was selected for its consistently strong performance across diverse datasets, robustness to overfitting, and ability to handle imbalanced data without extensive preprocessing. It also requires minimal parameter tuning and provides interpretable outputs through feature importance rankings, which are particularly valuable in clinical and biological contexts.

The diagnostic assist DLS was trained on a dataset constructed independently from the training and validation sets used for the image recognition DLS. This dataset comprised 39 features: 30 derived from image recognition DLS, including cell type and morphological characteristics with sensitivity, specificity, accuracy, and AUC all exceeding 0.8; and 9 from the Sysmex XN-9000 automated hematology analyzer, including red blood cell (RBC) count, WBC count, hemoglobin (Hb), hematocrit (HCT), mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), mean corpuscular hemoglobin concentration (MCHC), red cell distribution width (RDW), and platelet count (PLT). Each data entry was labeled with the corresponding diagnostic category. All diagnoses were confirmed by independent hematopathologists based on clinical information, laboratory data, flow cytometry analyses, genetic data, bone marrow aspiration and biopsy findings, in accordance with WHO criteria [24]. The WBC differential counts were excluded due to their redundancy with the DLS system and misalignment with microscopy classifications, such as segmented and band neutrophils, abnormal lymphocytes, and blast cells. Flags or interpretive program (IP) messages generated by the XN analyzer were not utilized because they are produced by a specific algorithm based on numerical data, particle size distributions, and scattergrams, rather than on the morphological features of blood cells.

To identify key features contributing to classification, hierarchical clustering was performed on the dataset. The dataset was

divided into five equal-sized subsets for k -fold cross-validation. In each round, four folds were used as training and one fold as validation. This process was repeated five times, allowing all cases to be used for both training and validation. It helps reduce overfitting and ensures a robust evaluation of model

performance by testing the model across multiple data splits [25–27].

To improve the interpretability of the diagnostic assist DLS in clinical applications, we incorporated SHAP (SHapley Additive

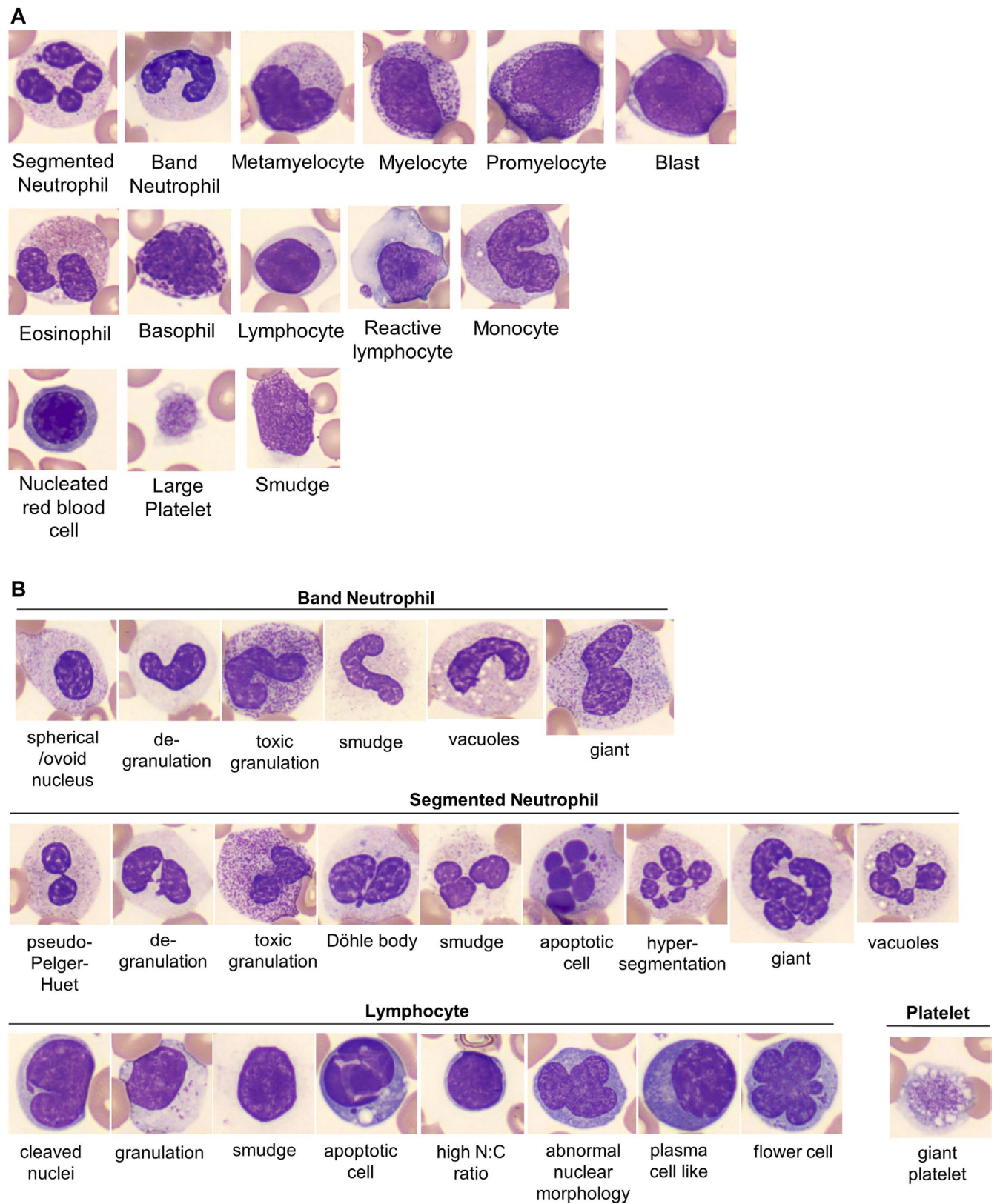


FIGURE 1 | Legend on next page.

FIGURE 1 | (A) Representative images of 14 types of differentiated blood cells. Granulocytes include segmented neutrophils (lobulated nuclei), band neutrophils (rod-shaped nuclei), eosinophils, and basophils. Immature myeloid precursors include promyelocytes, myelocytes, and metamyelocytes. Agranulocytes include lymphocytes, reactive lymphocytes, and monocytes. Abnormal or atypical findings include blasts, nucleated red blood cells (NRBCs), large platelets, and smudge cells. (B) Representative images of abnormal morphological characteristics. Band neutrophils: Spherical/ovoid nucleus—nucleus is rounded or oval-shaped rather than band-like; Degranulation—loss or reduction of cytoplasmic granules; Toxic granulation—presence of abnormally large, dark-staining granules; Smudge—disrupted cell with indistinct cytoplasmic borders and scattered nuclear material; Vacuoles—clear, round spaces within the cytoplasm; Giant—enlarged cell with increased nuclear or cytoplasmic size. Segmented neutrophils: Pseudo-Pelger-Huet—hypolobated nucleus with dense chromatin; Degranulation—reduced granularity in the cytoplasm; Toxic granulation—coarse, dark cytoplasmic granules; Döhle body—pale blue-gray oval inclusion in the cytoplasm; Smudge—fragmented or smeared nuclear or cytoplasmic material; Apoptotic cell—cell showing nuclear condensation and fragmentation; Hypersegmentation—nucleus with more than five lobes; Giant—significantly enlarged cell; Vacuoles—round clear cytoplasmic spaces. Lymphocytes: Cleaved nuclei—nucleus with deep indentations or clefts; Granulation—presence of visible cytoplasmic granules; Smudge—disrupted cell with diffuse nuclear material; Apoptotic cell—nuclear condensation and fragmentation indicating cell death; High N:C ratio—large nucleus relative to a small amount of cytoplasm; Abnormal nuclear morphology—irregular nuclear shape or chromatin pattern; Plasma cell-like—eccentric nucleus with adjacent perinuclear clearing; Flower cell—multilobed nucleus forming a petal-like appearance. Platelet: Giant platelet—abnormally large platelet, larger than a red blood cell.

Explanations) as an explainability tool. SHAP values were calculated for each feature to help clinicians and laboratory technicians identify the key factors influencing the model decisions [6, 28]. Additionally, RF prediction probabilities were projected into 3D coordinates to visualize the distribution of samples in the diagnostic space.

2.4 | Statistics

Data were analyzed for statistical significance using JMP Pro 17 and Python 3.11. Correlation analysis between the DLS and the hematopathologist was performed using Pearson's correlation coefficient. Analysis of variance (ANOVA) was employed to calculate significance across more than two groups. Any case with $p \leq 0.05$ was considered statistically significant. SHAP values were calculated using the SHAP library. Matplotlib and mpl_toolkits Mplot 3d were utilized for creating 3D plots. Clustering, AUC, and ROC curves were calculated using the scikit-learn library. Sensitivity, specificity, accuracy, as well as the ROC curve and AUC indices, were used to assess the performance of the machine learning model.

This study was approved by the Juntendo University Hospital Medical Ethics Committee (Tokyo, Japan) (IRB #H16-0251). As part of the approval, the ethics committee explicitly waived the need for informed consent from individual patients as all the samples were de-identified in accordance with the Declaration of Helsinki.

3 | Results

3.1 | Blood Cell Classification of the Image Recognition DLS

3.1.1 | Blood Cell Classification Performance of the Image Recognition DLS

Table 1 illustrates the cell classification accuracy, sensitivity, and specificity ranging from 97.3% to 99.9%, 66.5% to 99.8%, and 98.5% to 100%, respectively. Twelve of the 14 cell types showed an accuracy greater than 99%. These evaluation

TABLE 1 | Cell classification performance of the image recognition DLS in the evaluation dataset.

Cell type	Sensitivity (%)	Specificity (%)	Accuracy (%)
Segmented neutrophil	97.9	99.1	98.6
Band neutrophil	87.3	99.0	98.6
Metamyelocyte	66.5	99.9	99.8
Myelocyte	84.6	99.9	99.8
Promyelocyte	73.4	100	99.9
Blast	81.8	99.8	99.2
Eosinophil	99.3	100	99.9
Basophil	85.6	99.9	99.8
Lymphocyte	93.5	98.5	97.3
Reactive lymphocyte	84.1	98.9	98.8
Monocyte	97.8	99.8	99.6
Nucleated red blood cell	96.4	99.8	99.8
Large platelet	99.8	99.9	99.9
Smudge	84.8	99.7	98.8

metrics for segmented neutrophil, eosinophil, lymphocyte, monocyte, NRBC, and large platelet were higher than 95%. However, the sensitivity was lower than 80% for promyelocyte and metamyelocyte. Table 2 shows the performance of the image recognition DLS in detecting morphological anomalies as morphological characteristics. The sensitivity and specificity of 10 out of 14 morphological characteristics, as well as the accuracy of 11 out of 14 morphological characteristics, were greater than 90%.

As shown in Table 3, the image recognition DLS tended to misclassify cells belonging to the same lineage, such as myeloid

TABLE 2 | Morphological characteristics identification performance of the image recognition DLS in the evaluation dataset.

Morphological comment	Sensitivity (%)	Specificity (%)	Accuracy (%)
Band neutrophil			
Spherical/ovoid nucleus	100.0	88.4	88.7
Degranulation	82.5	96.3	96.1
Toxic granulation	94.6	94.8	94.8
Smudge	92.8	96.1	95.9
Segmented neutrophil			
Pseudo-Pelger-Huet	95.9	91.9	92.0
Degranulation	98.4	91.0	91.1
Toxic granulation	96.0	98.1	98.0
Döhle body	80.8	86.4	86.1
Smudge	99.0	94.9	95.0
Apoptotic cell	94.7	90.9	90.9
Lymphocyte			
Cleaved nuclei	94.4	81.4	81.6
Smudge	95.2	98.4	98.2
Apoptotic cell	96.7	95.3	95.3
Platelet			
Giant platelet	98.5	95.5	96.3

precursor cells at adjacent maturity levels. For example, segmented neutrophils were misclassified as band neutrophils, band neutrophils as metamyelocytes, metamyelocytes as myelocytes, and myelocytes as promyelocytes. The DLS encountered challenges in differentiating lymphocytes from variant lymphocytes. Some lymphoma cells were classified as blasts, and some plasma cell-like cells were misclassified as NRBCs.

The correlation between the image recognition DLS and the reference method by human eyes was examined for each digital cell image. All cell types showed strong correlations ($r \geq 0.900$), except for variant lymphocytes ($r = 0.553$) and promyelocytes ($r = 0.790$) (Figure S1).

3.1.2 | Identification of Blasts and Abnormal Cells

Of the 589 routine examination smears, the manual microscopic examination detected abnormal findings in 118 PB smears: blasts were detected in 109 smears and ML-like abnormal lymphocytes, including cleaved and plasma cell-like lymphocytes, were found in 11 smears. In two cases, blasts and abnormal lymphocytes were detected simultaneously. Regarding blast detection, the image recognition DLS correctly identified blasts in all 109 smears that were detected by the reference method, achieving sensitivity of 100%, specificity of 77.1%, and accuracy of 81.3%. Figure S2A displays blasts detected by the image recognition DLS among 17 AML cases and 8 ALL cases with blasts in the PB according to manual microscopy.

In contrast, we found that the image recognition DLS detected blasts in 16 out of 214 HC smears (7.5%) where manual microscopic examination did not detect any blasts (Figure S2). Expert review of the DLS results showed agreement on blast determination in 11 out of 16 smears (68.8%), where blasts were detected by the DLS. At the same time, there were 19 cells in these 16 smears that the expert reviewer could not determine as morphologically obvious blasts. The reviewers were unable to clearly classify these cells as blasts, lymphocytes, variant lymphocytes, monocytes, or myelocytes. Cells that were difficult to classify were determined by consultation among the three reviewers. Additionally, some images that the DLS incorrectly identified as blasts contained more than one cell in a single image area, suggesting that the DLS's automatic identification may not have worked properly. Handling images with multiple cells captured is a concern for future consideration.

Manual microscopic examination detected abnormal cells characteristic of mature lymphoid neoplasms in 12 smears, including those from adult T-cell leukemia/lymphoma (ATL), follicular lymphoma (FL), diffuse large B-cell lymphoma (DLBCL), mantle cell lymphoma (MCL), and plasma cell leukemia (PCL) cases (refer to Table 4). The image recognition DLS identified an increase in lymphocytes with accompanying morphological characteristics, as well as the presence of blasts in smears from mature lymphoid neoplasms, excluding PCL cases. In Figure S2C, lymphocytes with cleaved nuclei, abnormal nuclear morphology, apoptotic cells, plasma cell-like cells, and flower cells are identified as lymphoma cells. The number of lymphocytes with morphological characteristics detected by

TABLE 3 | The confusion matrix of the 14 types of differentiated blood cells is compared to the reference classification of the evaluation dataset.

DLS output	Reference												
	SNE	BNE	MMY	MY	PMY	BL	EO	BA	LY	VLY	MO	NRBC	LP
SNE	56357	494	7	1	0	2	4	66	2	0	18	1	0
BNE	1121	4141	60	5	0	1	4	12	11	0	12	3	0
MMY	10	36	270	11	0	2	0	2	4	0	10	1	0
MY	4	3	30	686	10	43	1	2	59	0	14	0	0
PMY	0	0	1	27	91	7	1	0	5	1	13	0	0
BL	0	0	1	10	9	3621	1	5	218	5	11	0	0
EO	14	6	0	3	0	6	3233	27	2	0	0	0	0
BA	13	4	5	16	4	12	5	1155	6	0	0	0	0
LY	6	9	9	28	1	541	3	5	29253	146	31	10	2
RLY	0	0	0	1	1	27	0	0	1351	1223	43	1	0
MO	27	35	15	13	7	44	0	0	22	78	7199	0	0
NRBC	0	1	1	1	0	8	0	0	179	2	0	1182	1
LP	0	0	0	0	0	2	0	0	0	0	0	1	6696
SMU	31	15	7	6	1	17	3	69	83	0	11	10	1
Total	57583	4744	406	808	124	4333	3255	1343	31195	1455	7362	1209	6700

Note: Panel color indicates the percentage within each column $\geq 50\%$, 10% – 50% , and 0% – 10% .

Abbreviations: BA, basophil; BL, blast; BNE, band neutrophil; EO, eosinophil; LP, large platelet; LY, lymphocyte; MMY, metamyelocyte; MO, monocyte; MY, myelocyte; NRBC, nucleated red blood cell; PMY, promyelocyte; SNE, segmented neutrophil; SMU, smudge; VLY, variant lymphocyte.

DLS in these smears was significantly higher than in HC smears (cleaved nuclei $p < 0.001$, flower cell $p < 0.001$, see Figure S3). A weak positive correlation was observed between cleaved nuclei and flower cells ($r = 0.236$, $p < 0.001$), while a moderate negative correlation was found between cleaved nuclei and smudge cells ($r = -0.458$, $p < 0.001$) in ML samples.

In the smear of the PCL case, the image recognition DLS did not detect any blasts, nor did it identify an increase in lymphocytes with morphological characteristics. However, the XN analyzer reported increased monocytes and flagged abnormal messages of “monocytosis?” for suspected monocytes and “blast?” for suspected blasts. In specimens in which monocytosis was flagged in the XN analyzer, but not observed in the DLS, abnormal lymphocytes may be present in the monocyte region of the XN scattergram.

3.2 | Clinical Performance Evaluation of the Diagnostic Assist DLS; Identification of Healthy Control, MDS, and Non-MDS

Finally, we evaluated the clinical performance of the diagnostic assist DLS, which combines data from the image recognition DLS with that of the XN analyzer, in differentiating between HC, MDS, and non-MDS cases. PB smears were analyzed from 213 HC individuals (excluding cases with chronic kidney disease), 45 MDS patients (excluding posttransplant cases and those with peripheral blasts), and 279 non-MDS patients.

As clustering result shows in Figure 2A, variation of morphological characteristics is rarely observed in the HC group but is frequently observed in the MDS and non-MDS groups. The diagnostic assist DLS yielded a higher frequency of giant platelets and SN-pseudo-Pelger-Huet for morphological characteristics, as well as MCV in the MDS group compared to the HC and non-MDS groups. BN and immature myeloid precursors (MMY, MY, and PMY), as well as SN-Döhle bodies, were significantly higher in the non-MDS group compared to the MDS group. Additionally, indicators of bone marrow stress or malignant cell infiltration, such as NRBCs, immature granulocytes, and blasts, were unevenly observed in the non-MDS group, while they were rarely seen in the MDS group. RBC, HGB, and Hct measured by XN analyzer and SN-neutrophil (%) detected by the DLS were lower in both the MDS and non-MDS groups compared to the HC group.

A strong positive correlation was observed between SHAP values and intrinsic feature importance from the RF model (Pearson's $r = 0.962$, $p < 0.001$), indicating consistent feature ranking across both interpretability methods.

The diagnostic assist DLS correctly identified 212 out of 213 HC samples (AUC 1.00), 32 out of 45 MDS samples (AUC 0.99), and 278 out of 279 non-MDS samples (AUC 1.00) (see Figure 2B).

Figure 2C shows a 3D plot of the prediction probabilities for all cases. Each case is plotted on a scale of 0–1 representing the probability of belonging to each group (HC, MDS, and non-MDS). A total of eight cases were misdiagnosed with a confidence level of 50% or greater: seven MDS cases were predicted as non-MDS,

and one MDS case was predicted as HC (refer to Table S2). Among the MDS cases misclassified as non-MDS were those that progressed to AML with myelodysplasia-related changes (AML-MRC) within a short period of time after the evaluation smear collection. Additionally, high-risk MDS (Pt 7–9) showed no blasts in the PB at the time of evaluation, but blasts appeared in the PB within days to weeks. Cases with abnormal immunologic mechanisms such as rheumatoid arthritis, ulcerative colitis with elevated plasma TPO, or non-hematologic malignancies were also predicted as non-MDS.

Of note, one MDS case misclassified as HC was later diagnosed with MDS with multisystem dysplasia (MDS-MLD) 1 year after this evaluation.

4 | Discussion

In this study, the performance of the enhanced image recognition DLS was evaluated using routine smears, demonstrating its ability to detect leukemia cases with blasts in PB, mature ML cases with malignant cell appearance, and MDS cases with morphological dysplasia through the combination of cell classification and identification of abnormal morphological features.

Regarding the detection of ML cells in PB, except for ML cells such as ATL and CLL that tend to proliferate in PB, most ML cells that infiltrate the bone marrow proliferate and subsequently enter the PB. Therefore, the detection of ML cells in PB is a significant and urgent finding indicating ML progression [29–32]. Although ML cell morphology may vary, they often display a notable uniformity, contrasting with the non-uniform morphology in reactive lymphocytosis triggered by viral infections. Recognizing this characteristic uniformity is essential for distinguishing ML cells from reactive lymphocytes [33]. This can be estimated by the percentage of lymphocytes given identical morphological characteristics by the image recognition DLS. The image recognition DLS does not directly identify lymphoma cells but rather identifies them as “abnormally shaped lymphocytes” within a broader category that may differentiate between normal and malignant lymphocytes within the same smear. A moderate negative correlation between cleaved nuclei and smudge cells in ML samples suggests that a higher proportion of smudged lymphocytes may obscure distinctive morphological characteristics that are important for lymphoma subclassification [34]. On the other hand, it is difficult to detect ML cells with poor morphological changes, and immunophenotyping together with morphological interpretation is considered essential for diagnosis [35, 36]. Nevertheless, this study demonstrates that the presence of ML cells can be identified by integrating the image recognition DLS with data from an automated hematology analyzer, even in PB smears where the DLS alone cannot detect specific ML cells. In this study, we were able to detect lymphoma cells in cases with lymphoid neoplasms that the DLS could not identify by combining CBC data with abnormal scattergrams. The combination of different testing methods, flow cytometry, and morphological DLS, offers valuable insights into detecting the suspected presence of an abnormal cell population. This is particularly useful when either method reveals significant abnormal findings or when there are discrepancies in the results obtained from both methods.

TABLE 4 | Characteristic features of abnormal lymphocytes observed in mature lymphoid neoplasms.

Diagnosis	Type of mature lymphoid neoplasms									
	ATL		DLBCL		MCL		PCL 1		PCL 2	
	Pt 1	Pt 2	Pt 3	Pt 4-1st	Pt 4-2nd	Pt 4-3rd	Pt 5-1st	Pt 5-2nd	Pt 6-1st	Pt 6-2nd
Routine examination results										
WBC ×10 ⁹ /L	12.0	6.3	7.6	30.7	29.9	45.3	16.2	21.2	16.6	11.3
Abnormal cells (%)	30.5 (cleaved lymphocytes)	10.0% (abnormal lymphoid cells)	7.5% (blastoid cells)	75.6% (abnormal lymphoid cells)	71.2% (abnormal lymphoid cells)	82.8% (abnormal lymphoid cells)	18.0% (abnormal lymphoid cells)	19.0% (blastoid cell)	95.0% (blastoid cell)	10.1% (plasma cell-like)
Image recognition DLS results										
Cell images (#)	223	218	204	213	249	242	243	240	239	214
Cell classification (#)										
Lymphocytes	72	89	6	117	136	109	34	37	42	35
Blasts	6	6	3	32	17	39	19	23	14	0
LY-morphological characteristic (%) ^a										
Cleaved nuclei	52.8	40.4	66.7	66.7	61.8	34.9	11.8	24.3	16.7	11.4
Flower cell	11.1	2.2	0	3.4	3.7	0.9	0	2.7	0	0
Plasma cell-like	0	0	0	0	0	0	2.9	0	0	5.7
Granulation	15.3	10.1	33.3	6	14	11.9	32.4	18.9	35.7	22.9
Smudge	8.3	13.5	0	3.4	16.2	2.8	26.5	16.2	19	11.4
Apoptotic cell	2.8	2.2	0	1.7	2.2	0.9	5.9	2.7	9.5	37.1

Abbreviations: ATL, adult T cell leukemia; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; MCL, mantle cell lymphoma; PCL, plasma cell leukemia.
^aLY-morphological characteristic (%); count of lymphocytes with morphological characteristics (DLS)/count of lymphocytes (DLS).

The image recognition DLS did not miss smears with blasts during routine examination, whether AML or ALL, achieving a sensitivity of 100%. This performance is comparable to or better than the detection rates of blasts reported in other studies

[33, 37]. On the other hand, the DLS detected blasts on several smears that had tested negative for blasts by the routine manual microscopic examination. This difference in blast detection between the DLS and manual microscopy may arise from

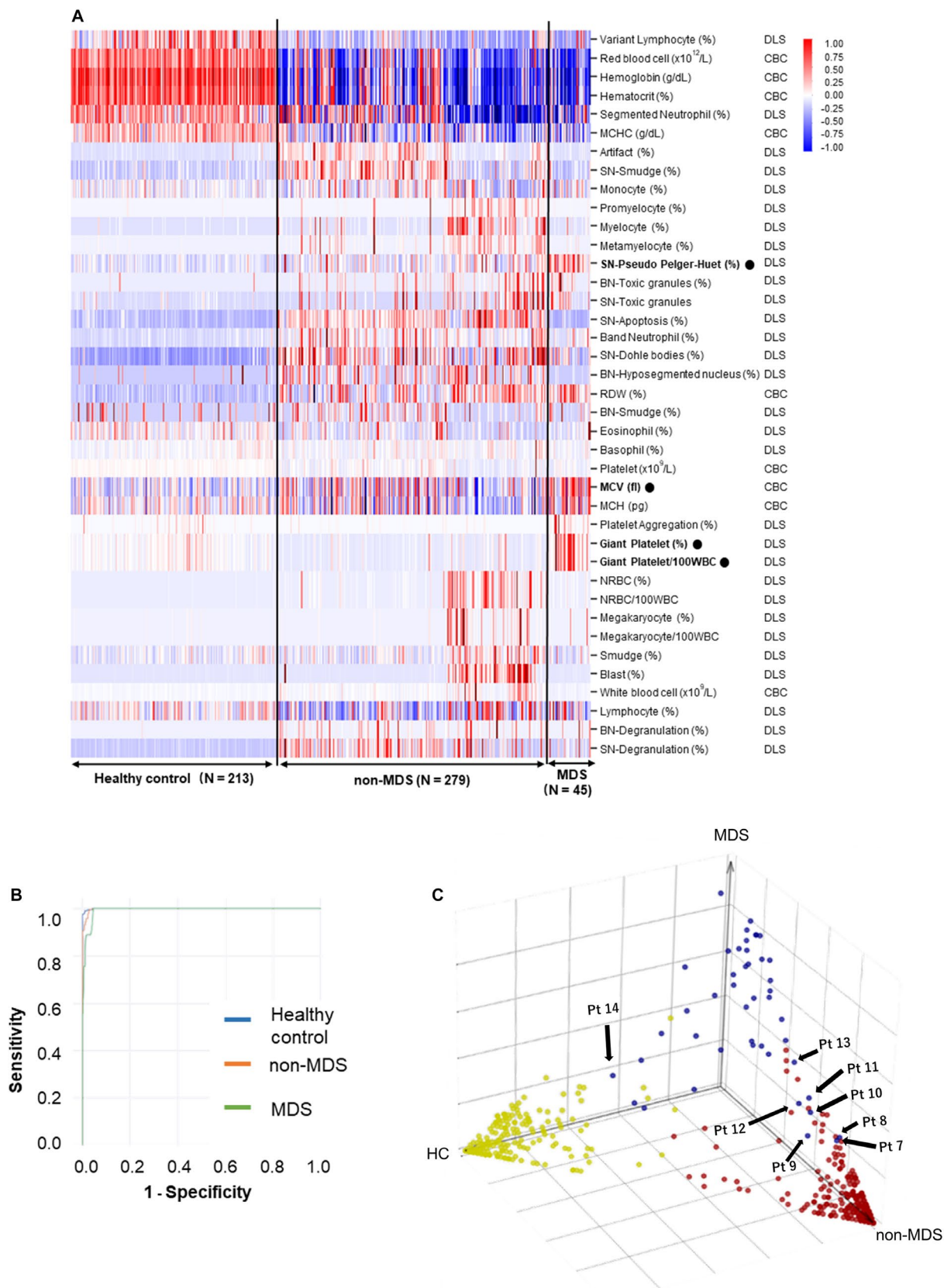


FIGURE 2 | Legend on next page.

FIGURE 2 | Parameters utilized in diagnostic assist DLS and its diagnostic performance. (A) A clustering map of normalized prediction value from machine learning, where rows represent distinct cell categories and features as identified by the DLS and CBC (Sysmex XN-9000) test outcomes. The individual samples used in this study are denoted along the columns. It revealed different patterns in the SHAP values across the three groups. (B) Differential diagnostic performance of the diagnostic assist DLS. ROC curve analysis for determining the healthy control (AUD 1.00), MDS (AUC 0.99), and non-MDS (AUC 0.99) cases. The AUC was used to measure performance; the maximum value is 1. (C) Three-dimensional plot of the prediction probabilities for all cases. Each case is plotted on a 0–1 scale as a probability function for each group (HC, MDS, and non-MDS). A total of eight cases were misdiagnosed with a confidence level of 50% or higher, and the details of each case are presented in Table S2. AUC, area under the ROC curve; BN, band neutrophil; DLS, deep learning system; MCH, mean corpuscular hemoglobin; MCHC, mean corpuscular hemoglobin concentration; MCV, mean corpuscular volume; RDW, red cell distribution width; ROC, receiver operating characteristic; SHAP, Shapley Additive exPlanations; SNE, segmented neutrophil.

variations in judgment criteria. The DLS relies solely on morphological analysis of digital images to detect blasts, while manual microscopy involves the synthesis of multiple factors. In manual microscopy, if a cell with blast-like morphology is identified, the smear is further scrutinized for the presence of additional similar cells. Subsequently, if two or more blasts are observed, or if clinical factors such as the patient's medical history, ongoing treatment, or CBC data suggest the possibility of blast presence, the smear is reported as blast-positive [38, 39]. False-positive blasts represent a weakness when integrating image recognition DLS into routine blood morphology testing. Nevertheless, leveraging the image recognition DLS to identify cells suspected to be possible blasts, even beyond the expertise of skilled laboratory technicians, holds value. If needed, these suspected cells can undergo review by a hematopathologist. Moreover, the utilization of deep neural networks by the DLS for blast identification with reproducible criteria can help standardize morphological blood cell examination, addressing a current challenge. When a single blast-like cell is detected in a smear, confirming subsequent changes over time by the image recognition DLS is a challenge to be addressed in the future.

The diagnostic assist DLS, which integrates image recognition DLS and quantitative data from the automated hematology analyzer, achieves high accuracy in differentiating MDS and non-MDS diseases. In this study, misclassification between MDS and non-MDS may have occurred because the non-MDS group included hematological malignancies, miscellaneous hematologic, and non-hematologic diseases. The diagnostic assist DLS misclassified MDS cases that progressed to AML-MRC within a short period of time after testing as non-MDS. This system may have the ability not only to differentiate MDS but also to predict the risk of progression from MDS to AML-MRC more accurately than assessment by the PB blast ratio alone.

However, the diagnostic assist DLS may be affected by various abnormal parameters. For instance, cases of anemia with lower MCV than non-MDS cases, as well as MDS specimens with comorbid chronic inflammatory diseases such as rheumatoid arthritis or cancer, might have been misclassified as non-MDS. Typically, MDS presents with high MCV and tends to exhibit macrocytic anemia [40–42]. In contrast, chronic inflammatory diseases often result in microcytic anemia due to impaired iron utilization [42, 43]. Cases of MDS complicated by chronic inflammatory diseases or malignancies may be affected by multiple parameter abnormalities due to a mixture of conflicting findings, making automated classification more challenging. These findings highlight the need for integrating additional clinical information, such as medical history, disease name,

previous results, and other laboratory data into the diagnostic workflow to improve accuracy and clinical relevance.

In this context, model interpretability plays a crucial role. SHAP-based feature rankings were highly consistent with the intrinsic feature importance derived from the RF model, particularly among the top predictive features, reinforcing the reliability and interpretability of the diagnostic system. In the future, we aim to develop an advanced diagnostic support system that combines hematologic image recognition, laboratory parameters, and clinical data to enhance diagnostic accuracy and support individualized clinical decision-making.

There are several limitations to this study. First, the number of cases of ML and MDS is limited. Second, there are issues with the automatic identification of digital images, such as the presence of multiple cells in one image area and the processing of blurred images. Additionally, staining conditions may affect the performance of the DLS; it is recommended to follow the manufacturer's recommended method when using the DLS in a clinical setting [23]. Third, the training and validation datasets for metamyelocytes and promyelocytes are relatively small compared to other cell types. While these counts are lower, they are sufficient for reliable statistical analysis based on established machine learning practices [44]. However, increasing the number of these images in future studies would further enhance the robustness of the analysis. Fourth, the scope of morphological characteristics incorporated into the DLS was limited due to insufficient data for training and validation. Finally, the current system was developed and validated using internal limited datasets; external validation using independent datasets from other institutions and clinical settings will be essential to confirm the generalizability, reproducibility, and clinical utility of the diagnostic assist DLS prior to broader implementation. Efforts toward this validation are currently underway.

In conclusion, this study demonstrates that a DLS trained with a sufficient number of high-quality labeled images and combined with data from an automated hematology analyzer can serve as a valuable diagnostic assistance tool. This integrated approach offers potential labor savings in the diagnostic process and promotes standardization of morphological testing. While CBC provides quantitative parameters, the DLS offers morphological analysis. The DI-60, available on the market, is used specifically for normal specimens, but incorporating the DLS, with its high capability to detect morphological abnormalities, will broaden its applicability in clinical settings. This integration may enhance laboratory workflows. For inexperienced technologists or facilities without a hematopathologist, the DLS may assist in

determining whether referral to a hematopathologist is necessary when analyzing PB smears. Our findings highlight the effectiveness of the DLS as a valuable tool in supporting diagnosis. We propose the implementation of a new “DLS-based diagnostic assist system,” which integrates morphological DLS and automated hematology analyzer data.

Author Contributions

Y.T. and Y.H. designed the study. S.N., J.B., Y.I., and M.R. carried out the experiments and performed the data analysis. Y.H., Y.T., and J.B. wrote the manuscript. M.A. and J.A. provided the idea for material preparation. A.M. and K.K. contributed to sample and data preparation.

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Ethics Statement

The research related to human use has been compiled with all the relevant national regulations, institutional policies, and in accordance with the tenets of the Declaration of Helsinki and has been approved by the Institutional Review Board (IRB) at Juntendo University Hospital (IRB #H16-0251).

Consent

Individual patient informed consents were waived as all samples were de-identified in accordance with the Declaration of Helsinki.

Conflicts of Interest

The Department of Advanced Research Institute for Health Science, Juntendo University Graduate School of Medicine has been endowed by Sysmex (Kobe, Japan) to develop and validate new diagnostic technology and to conduct academic research in hematology through collaborations. Y.I. and A.K. are employees of Sysmex. The study was performed by scientifically proper methods without any bias.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Correlation analysis between DLS and pathologists' classification of peripheral blood cells. Scatter plots depict linear regression lines showing the percentage of blood cell types after paired counting of 589 PB smears. % of Segmented neutrophil (95% CI [0.993–0.995]), Band neutrophil (95% CI [0.973–0.980]), Metamyelocyte (95% CI [0.872–0.906]), Myelocyte (95% CI [0.910–0.934]), Promyelocyte (95% CI [0.758, 0.819]), Blast (95% CI [0.964–0.974]), Eosinophil (95% CI [0.989–0.992]), Basophil (95% CI [0.939–0.956]), Lymphocyte (95% CI [0.952–0.965]), Variant lymphocyte (95% CI [0.495–0.607]), Monocyte (95% CI [0.968–0.976]), Nucleated red blood cell (95% CI [0.917–0.939]), Large platelet (95% CI [0.995–0.996]), and Smudge (95% CI [0.915–0.938]). **Figure S2:** (A) Blasts detected by the image recognition DLS among all 17 AML and 8 ALL cases with blasts in the peripheral blood by manual microscopy. One cell from each slide, showing representative blasts. (B) Blasts detected by the image recognition DLS among healthy control (HC) cases. (C) Blasts and lymphocytes with morphological characteristics among all mature lymphoid neoplasm cases with abnormal lymphoid cells in the peripheral blood by manual microscopy ALL, acute lymphoblastic leukemia; AML, acute myeloid leukemia; ATL, adult T-cell leukemia/lymphoma; DLBCL, diffuse large B-cell lymphoma; FL, follicular lymphoma; MCL, mantle cell lymphoma; PCL, plasma cell leukemia. **Figure S3:** Comparison of the dysmorphic features of the lymphocytes given by DLS among healthy controls (HCs) and patients with mature lymphoid neoplasms (only ATL, FL, DLBCL, and MCL cases). The vertical axis represents the rate of occurrence of LY-morphological characteristic (%) (count of lymphocytes with morphological characteristic given by the DLS/count of lymphocytes by the DLS). **Data S1:** Supporting Information.